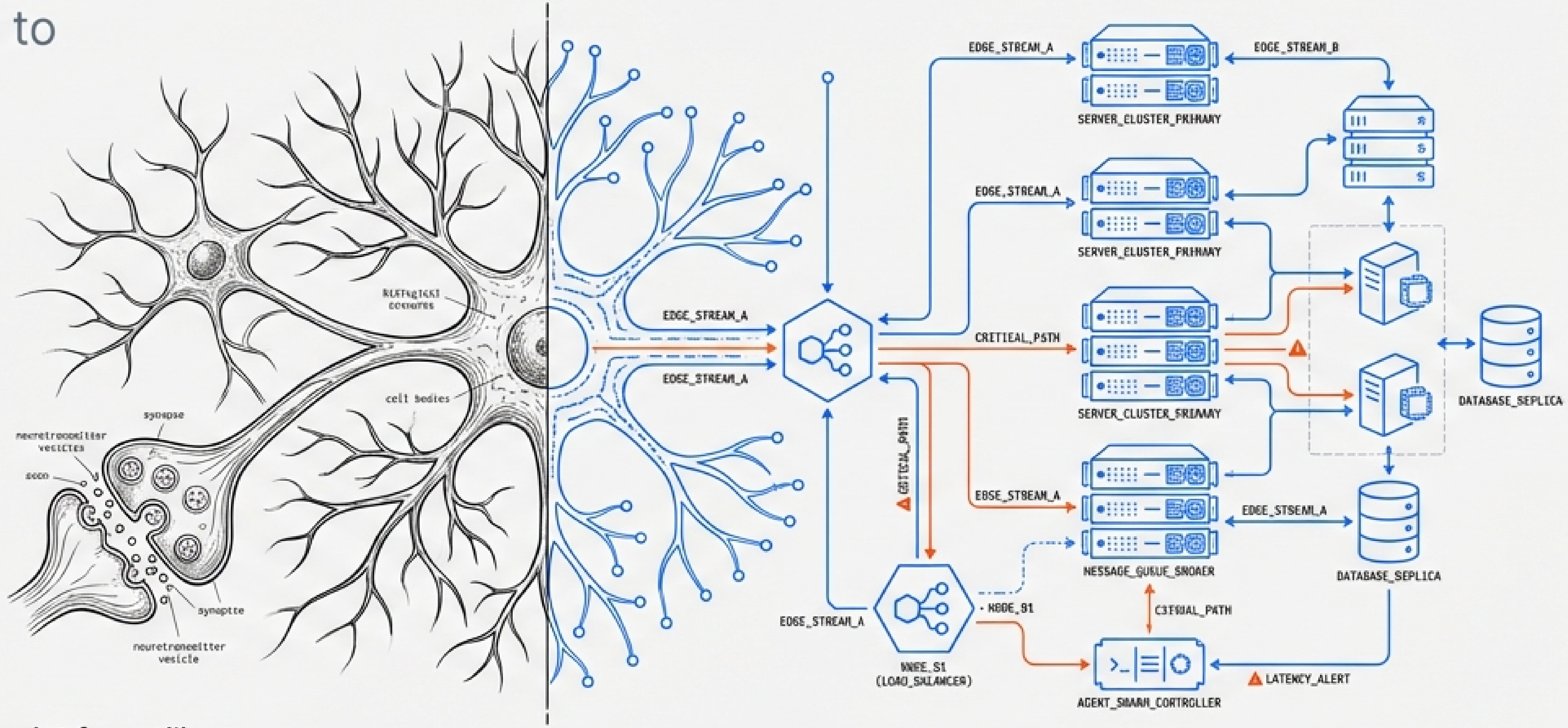
DEEP DIVE: THE SAMAS ARCHITECTURE

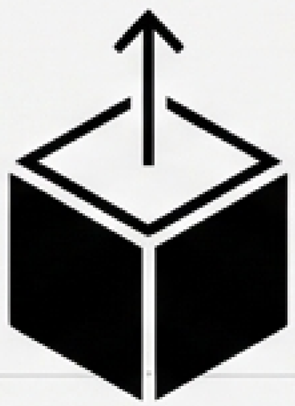
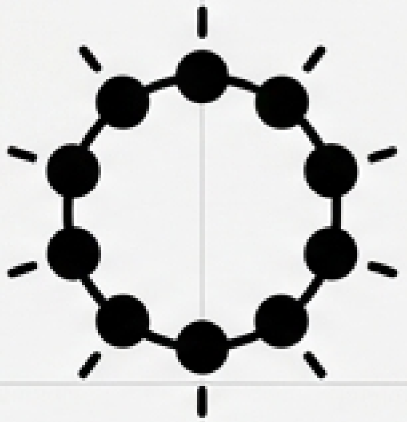
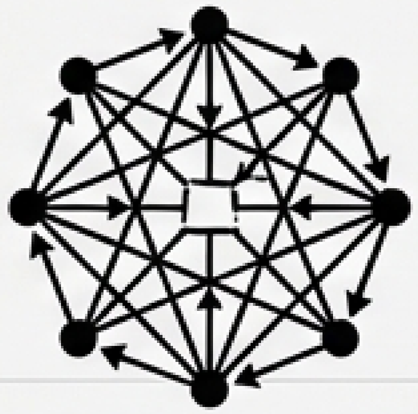
From Hobbyist Scripts to Superior Autonomous Multi-Agent Systems.



A comprehensive architectural blueprint for resilient, event-driven, and cost-optimized agent swarms. Based on forensic analysis of OpenClaw, SiteGPT, and the ‘Six Critical Unknowns’ of production deployment.

SYSTEM_STATUS: ARCHITECTURAL_REVIEW
VERSION: 1.0 (PRODUCTION_READY)
FOCUS: RESILIENCE // COST // SECURITY

THE EVOLUTION OF AGENCY

OPENCLAW	SITEGPT	SAMAS
The Local Monolith	The Polling Squad	The Enterprise Swarm
		
Architecture: Single Node.js Gateway Process	**Architecture:** 14 Specialized Agents on VPS	**Architecture:** Event-Driven (NATS JetStream)
Memory: File-based (MEMORY.md / SQLite)	**Coordination:** Shared Blackboard (Convex DB)	**Coordination:** Push-based “Chime-In” Protocol
Execution: Serial Lane Queue	**Execution:** 15-minute Polling Loop	**Execution:** Real-time K8s Autoscaling
Critical Flaw: Single point of failure. Process crash = System death.	**Critical Flaw:** High latency. Passive database, not active broker.	**Key Advantage:** Zero-Trust (WASM), Hybrid Memory, Resilience.

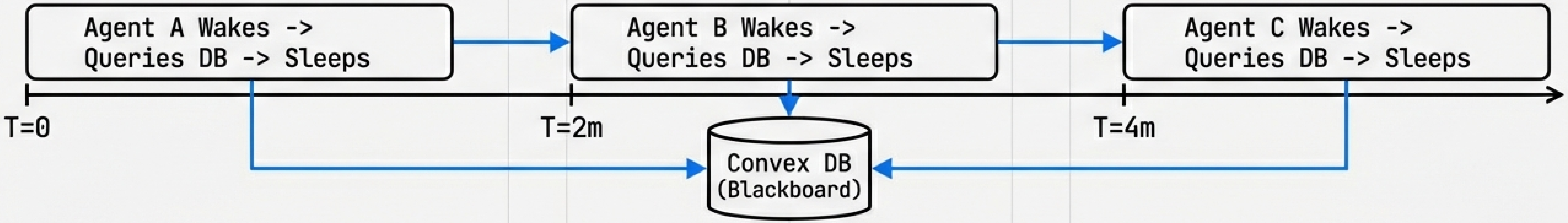
CRITICAL UNKNOWN #1: THE 'MISSION CONTROL' ARCHITECTURE

THE MYTH: Complex Websocket Mesh

Continuous, persistent, high-resource websocket mesh with unclear boundaries.

MYTH vs. REALITY

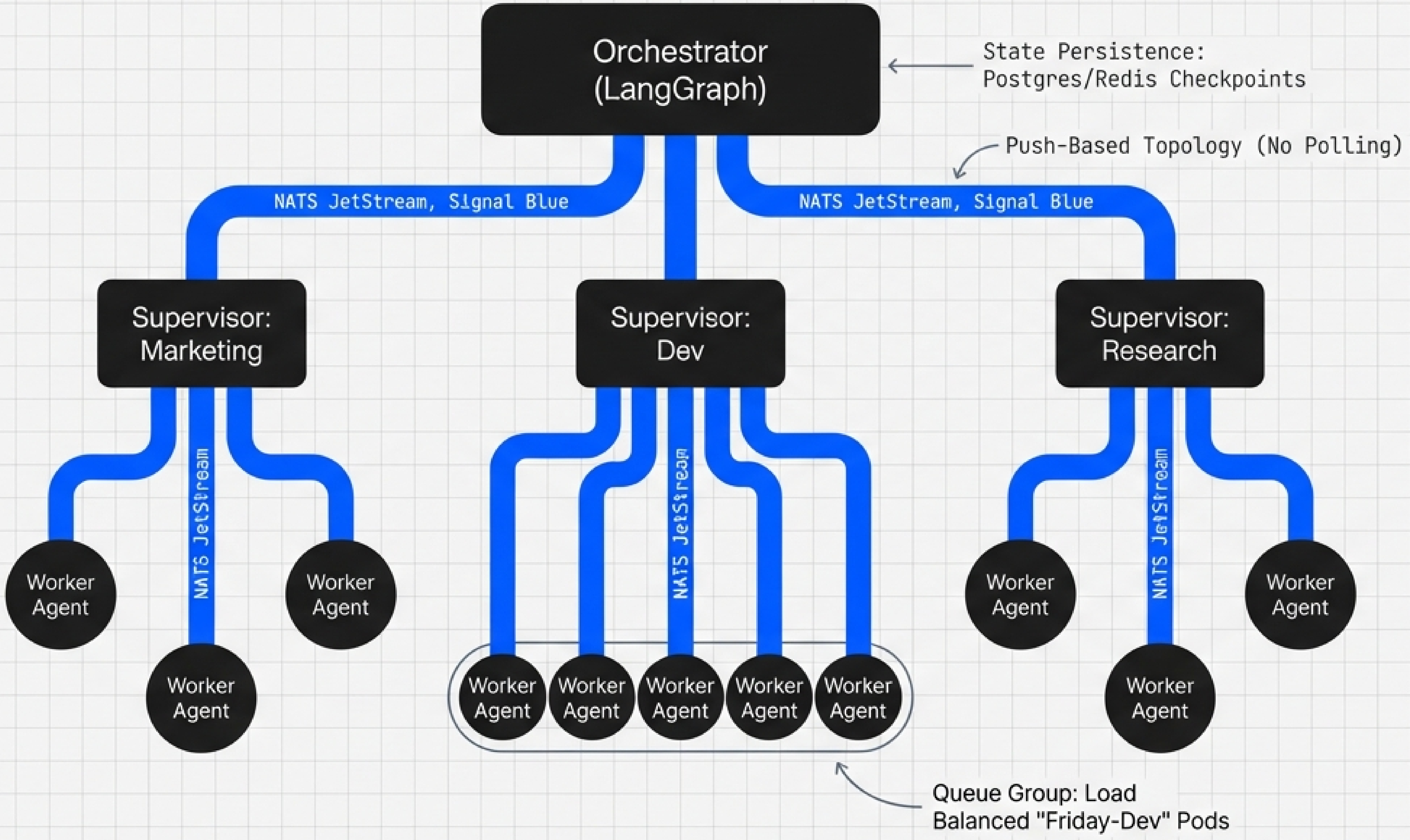
THE REALITY: Staggered Cron + Shared DB



KEY INSIGHT: Simplicity Scales. 10 Agents = 10 Isolated Sessions. Staggered execution prevents rate-limits and reduces compute costs to near zero when idle.

CRITICAL UNKNOWN #2: ORCHESTRATION & THE NERVOUS SYSTEM

Hierarchical Supervisor Pattern



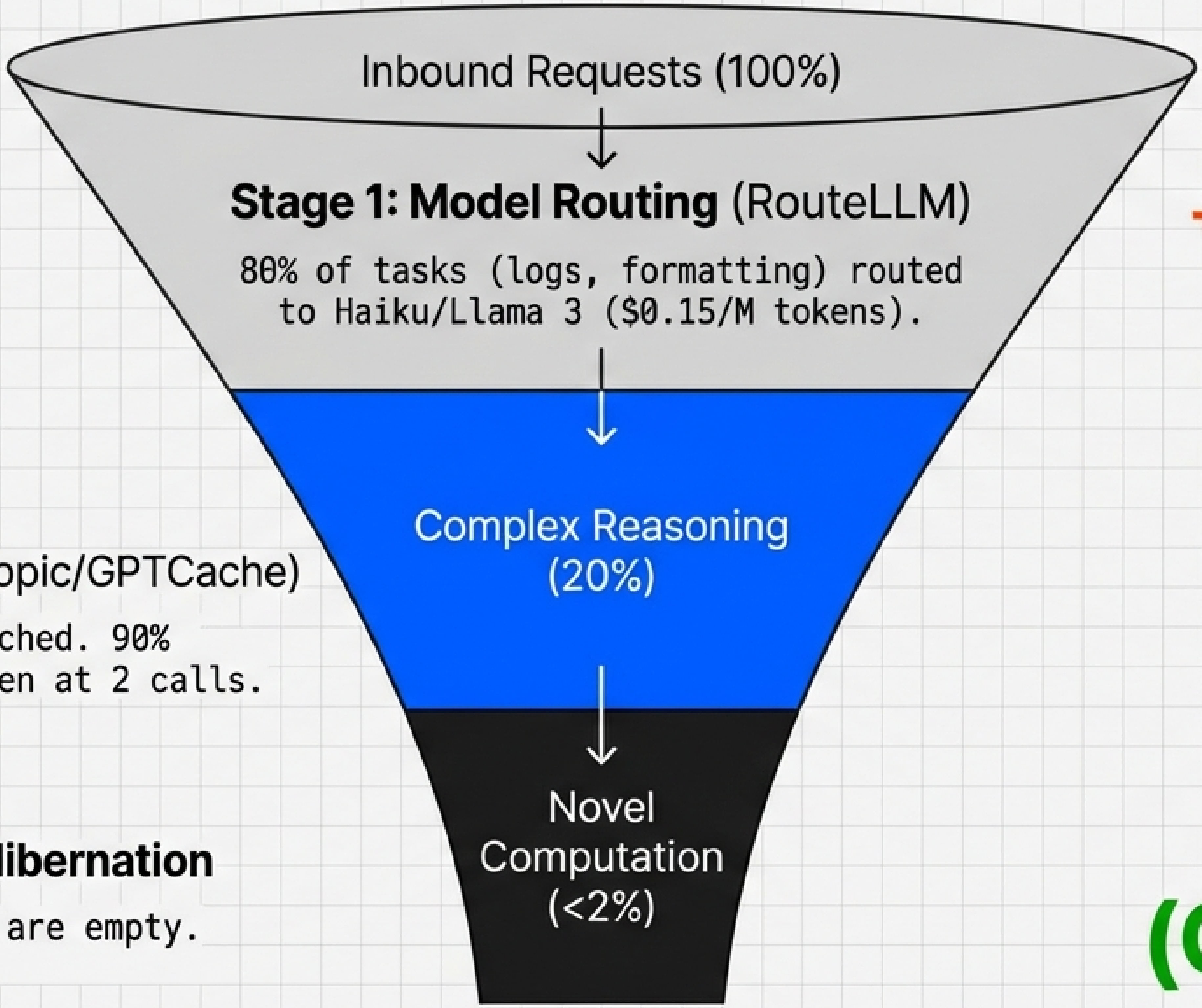
Framework: LangGraph

Chosen for cyclic graph support (loops) and "Pause/Resume" capability.

Backbone: NATS JetStream

Features "Interest-based retention"—tasks exist only while consumers exist.

COST OPTIMIZATION: THE $50k TO $500 FUNNEL



~~$50,000/mo~~
(Naive)

Stage 2: Caching (Anthropic/GPTCache)

System Prompts & Tools Cached. 90% Savings on Hits. Break-even at 2 calls.

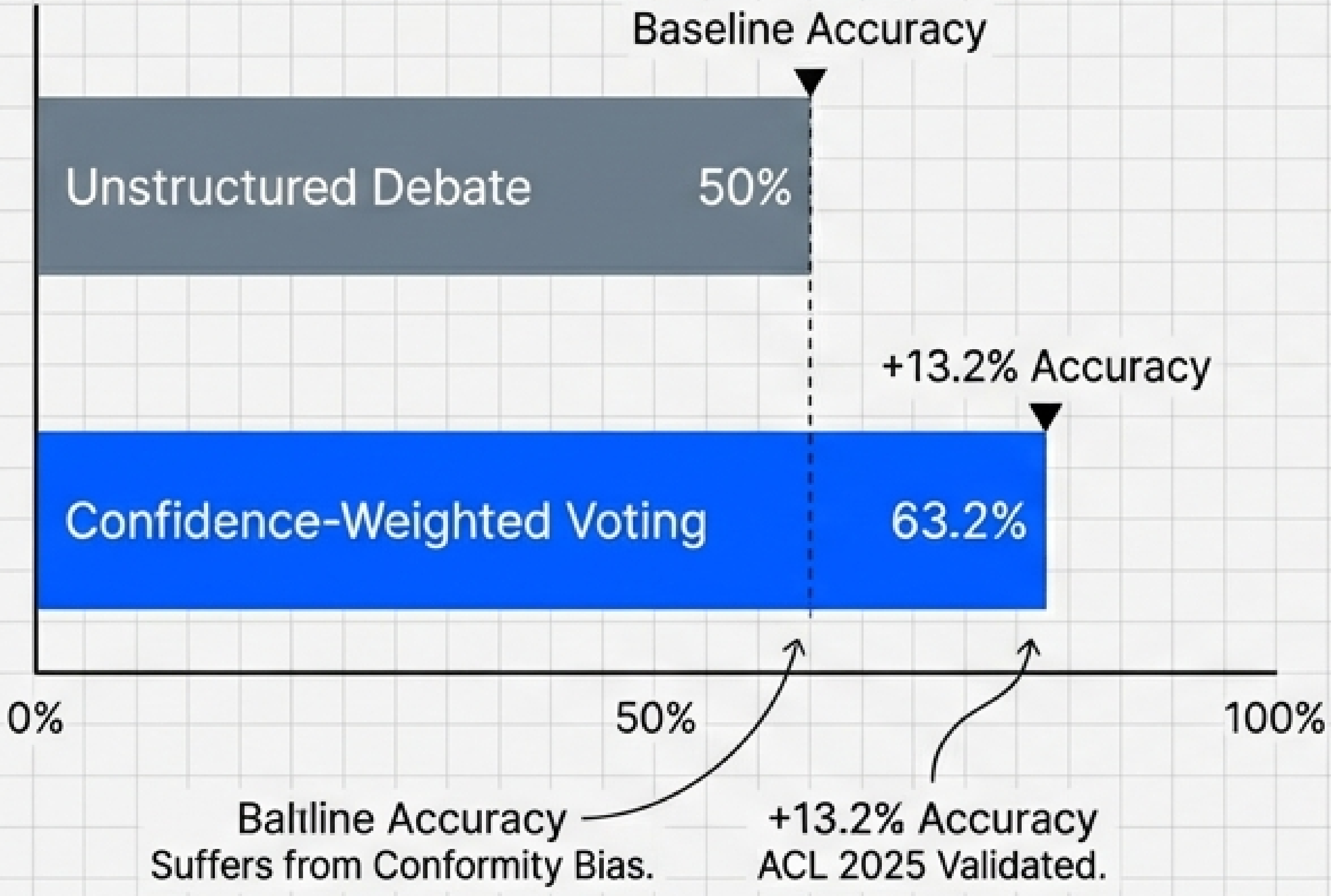
Stage 3: Event-Driven Hibernation

Agents sleep when queues are empty. Zero idle cost.

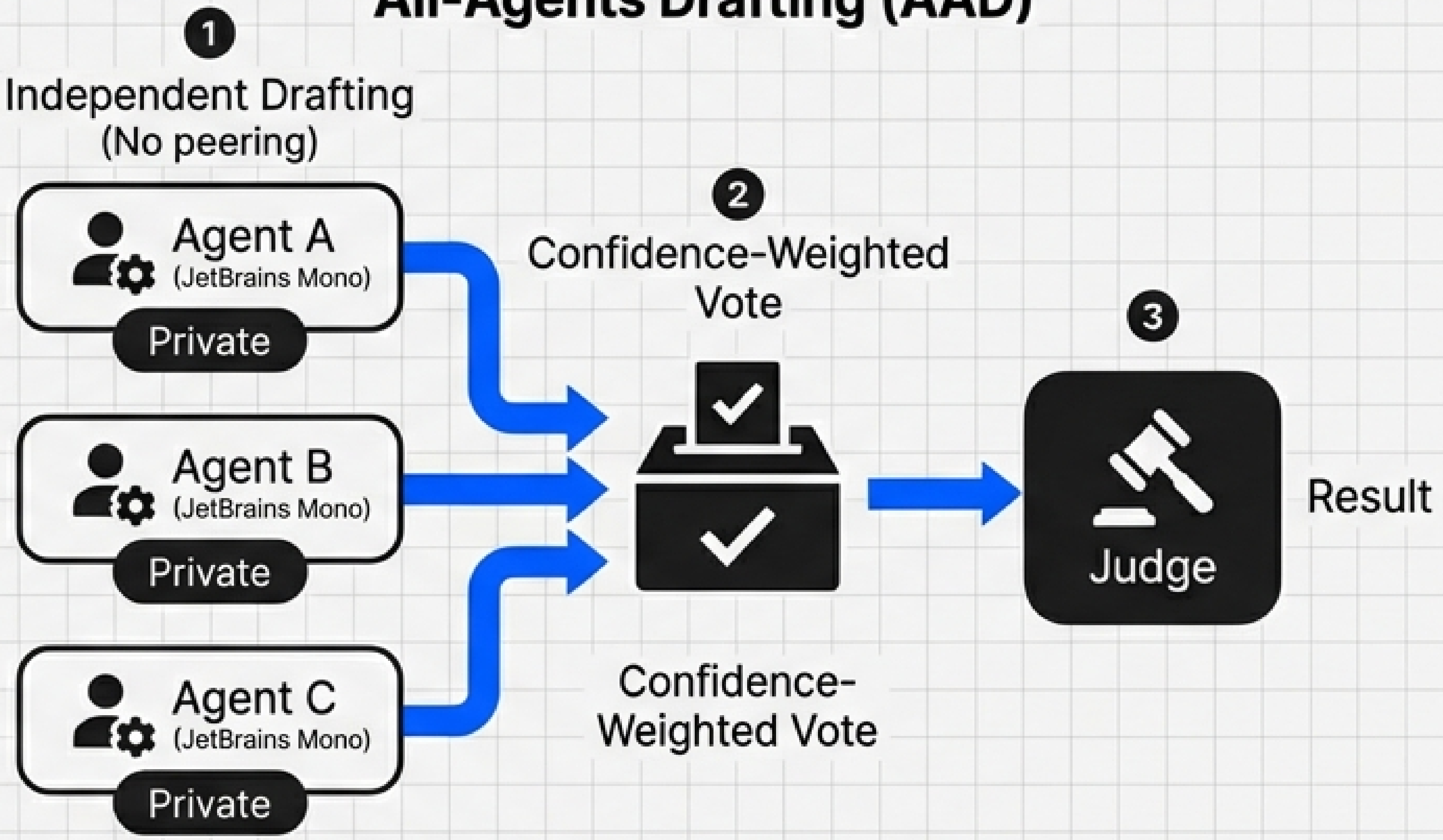
$500/mo
(Optimized)

CONSENSUS PROTOCOLS: TRUTH > DEBATE

Comparative Accuracy of Consensus Methods



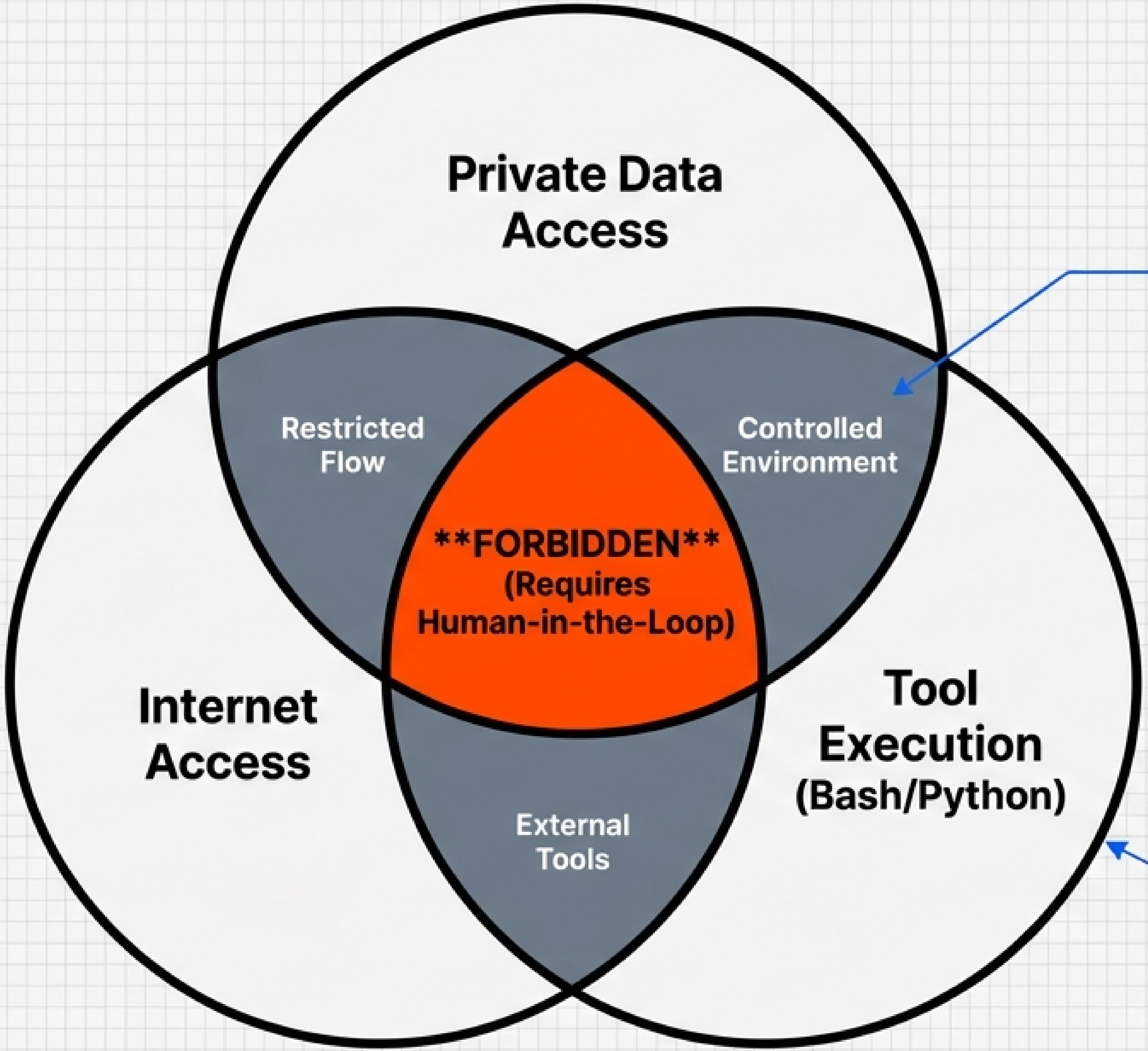
All-Agents Drafting (AAD)



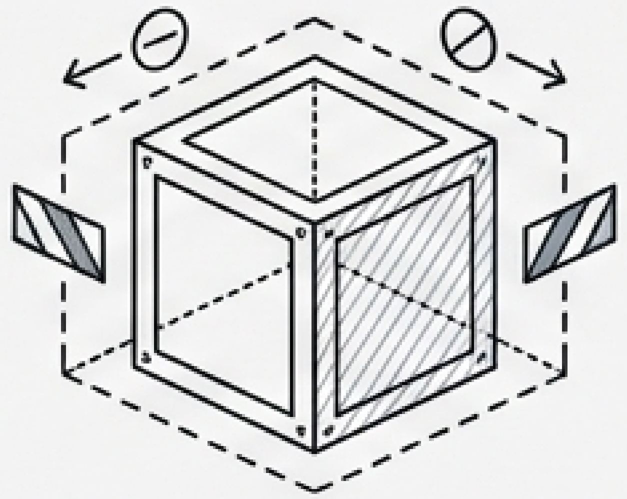
Key Insight:
Research shows that extended debate often *decreases* accuracy as agents conform to the majority view.

The **Solution**: Independent drafting followed by a Confidence-Weighted Vote.

SECURITY: THE "RULE OF TWO" & SANDBOXING



The Sandbox



`WASM / Firecracker MicroVMs`

- Hardware-level isolation.
- Millisecond startup.
- Prevents kernel escapes.

Defense in Depth

Coordinator + Guard Pattern: Independent sanitization and validation agents.

Provenance Ledgers: Tracking data origin to prevent prompt injection.

MEMORY ARCHITECTURE: THE HYBRID GRAPH

HOT MEMORY: Redis

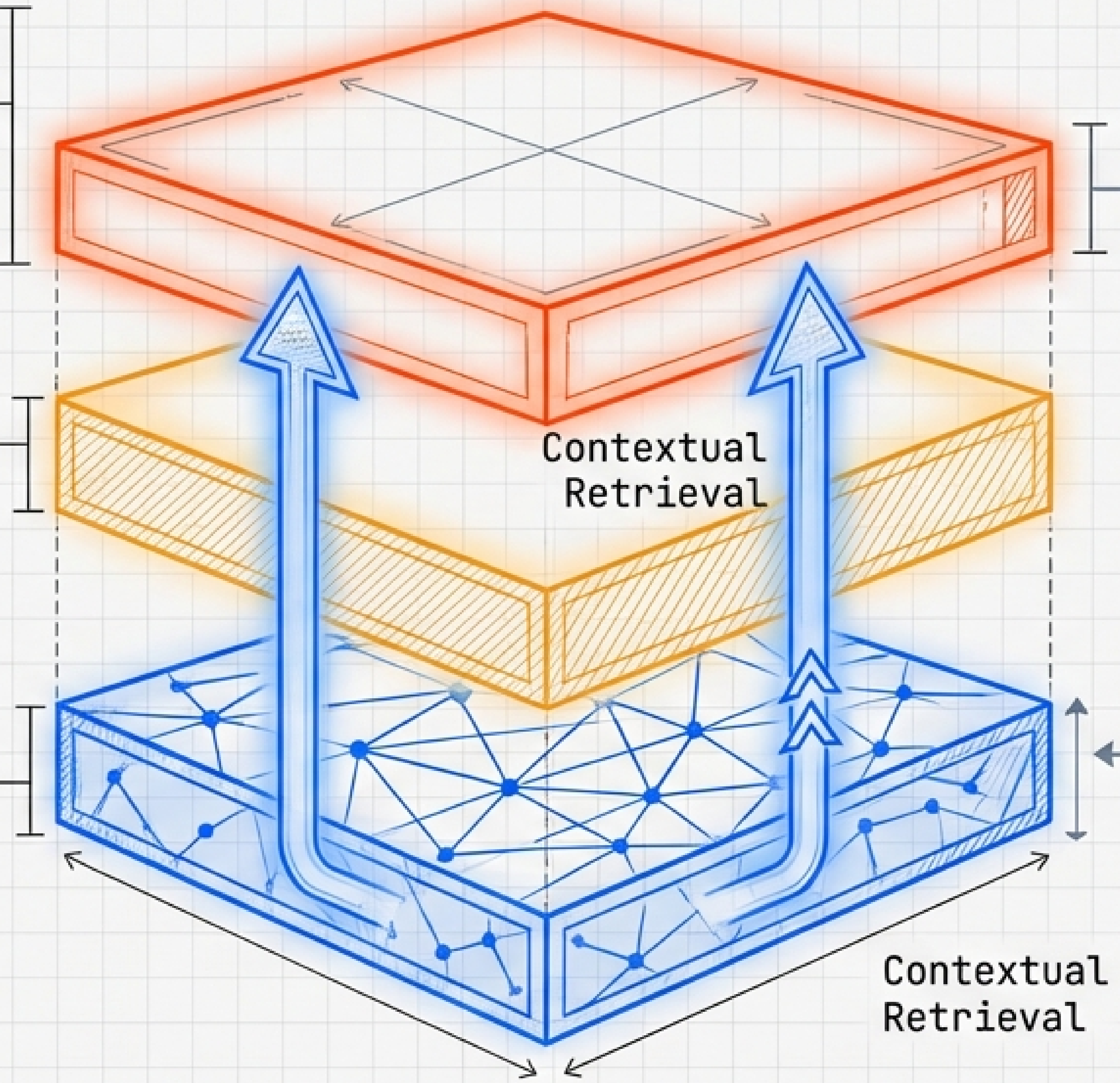
Thread-level persistence.
LangGraph Checkpointer.
Active task state.

WARM MEMORY: Qdrant

Vector/Semantic retrieval.
Payload-based partitioning for multi-tenancy.

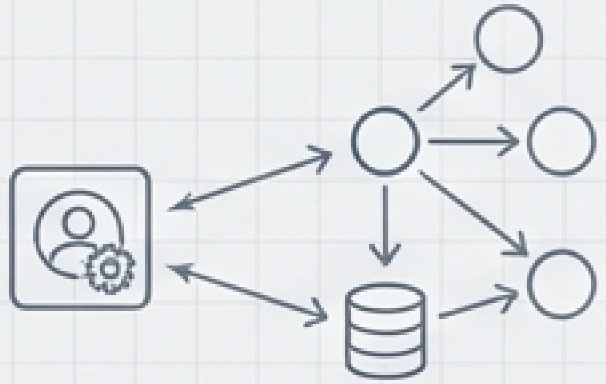
COLD MEMORY: Neo4j (GraphRAG)

Knowledge Graph. Connects disconnected facts (e.g., SEO Strategy <-> Q3 Financials).



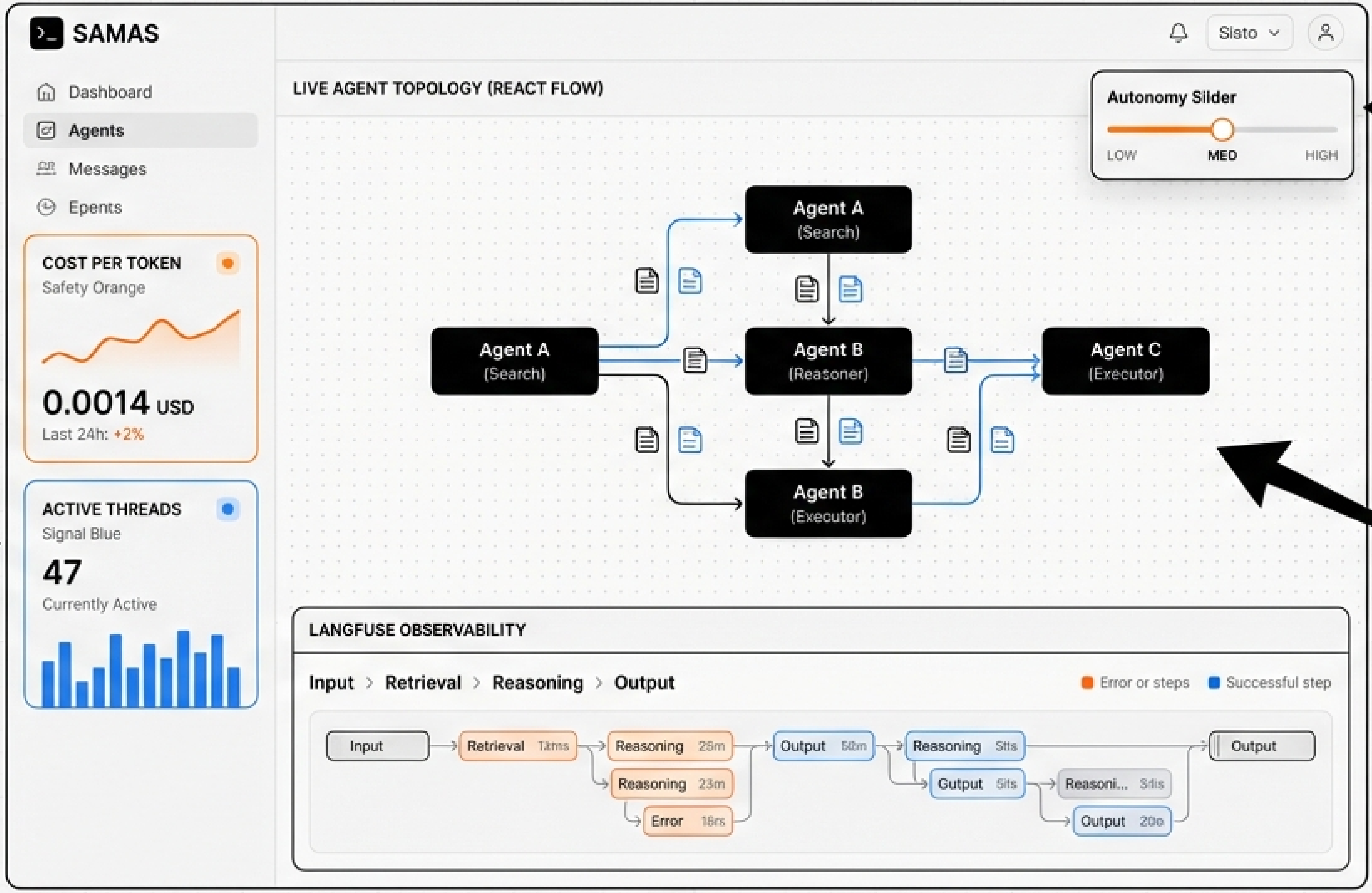
Technique Note:

Curator Agent asynchronously extracts entities (Nodes) and relationships (Edges) to populate the Graph.



THE DASHBOARD STACK: VISIBILITY & CONTROL

METRICS (TREMOR)



LANGFUSE OBSERVABILITY

Autonomy Slider: Low/Med/High settings per agent.

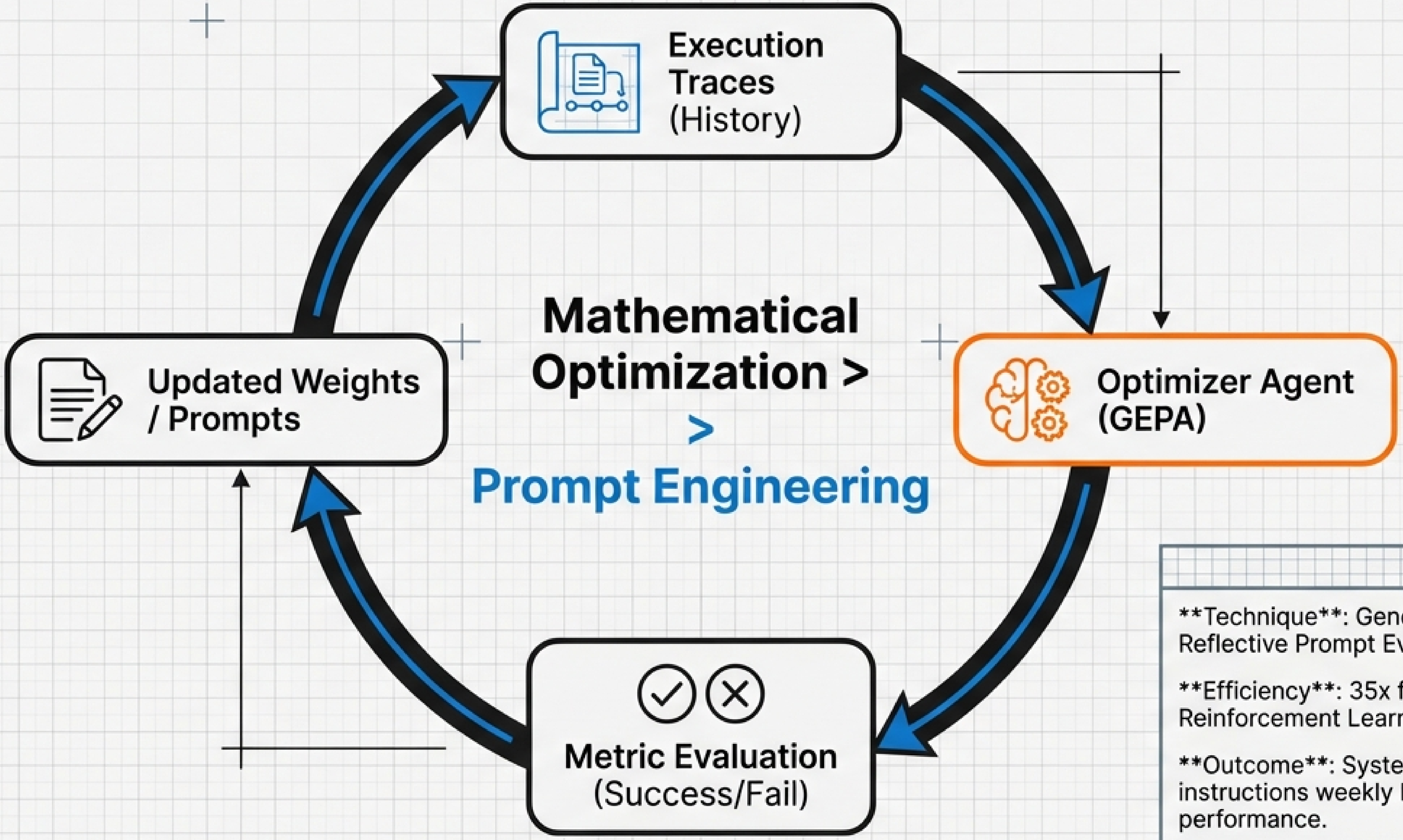
React Flow: Live Agent Topology Visualization.

Tremor: Real-time Cost Tracking.

```
Stack:
Next.js 16
+ shadcn/ui
+ Langfuse
```

NotebookLM

SELF-EVOLUTION: THE DSPy OPTIMIZATION LOOP



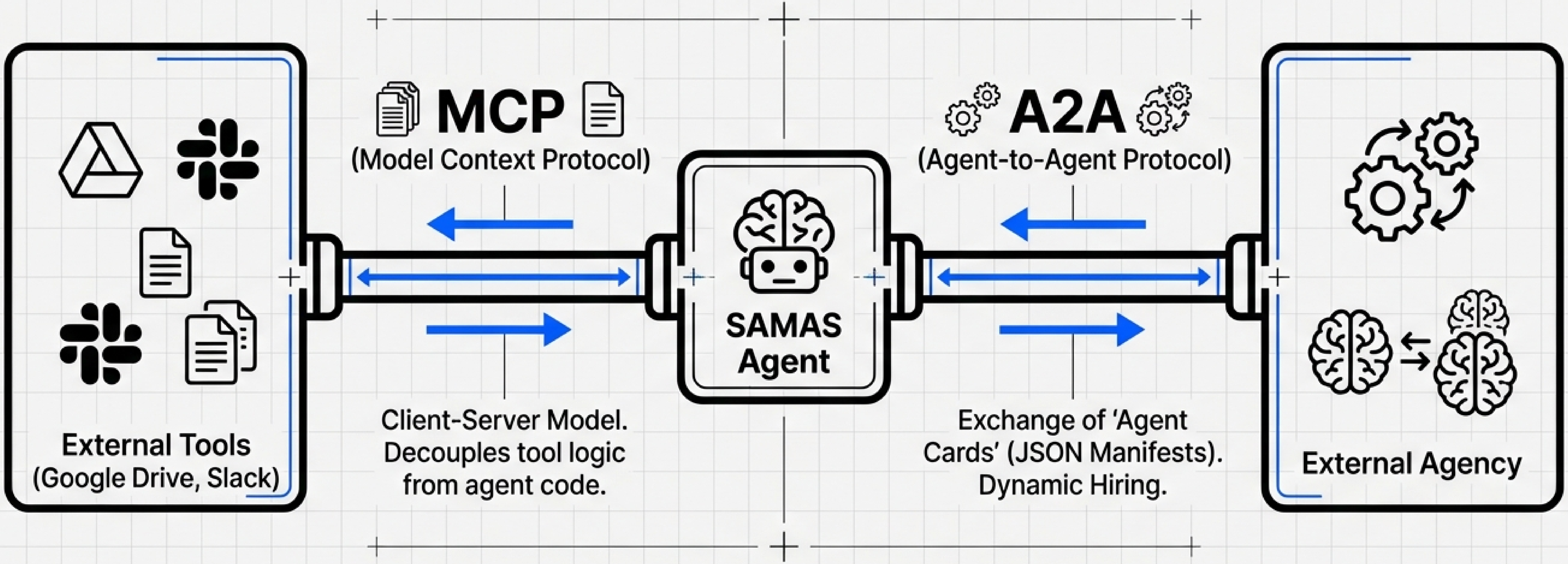
JetBrains Nono

Technique: Genetic-Pareto Reflective Prompt Evolution (GEPA).

Efficiency: 35x fewer rollouts than Reinforcement Learning.

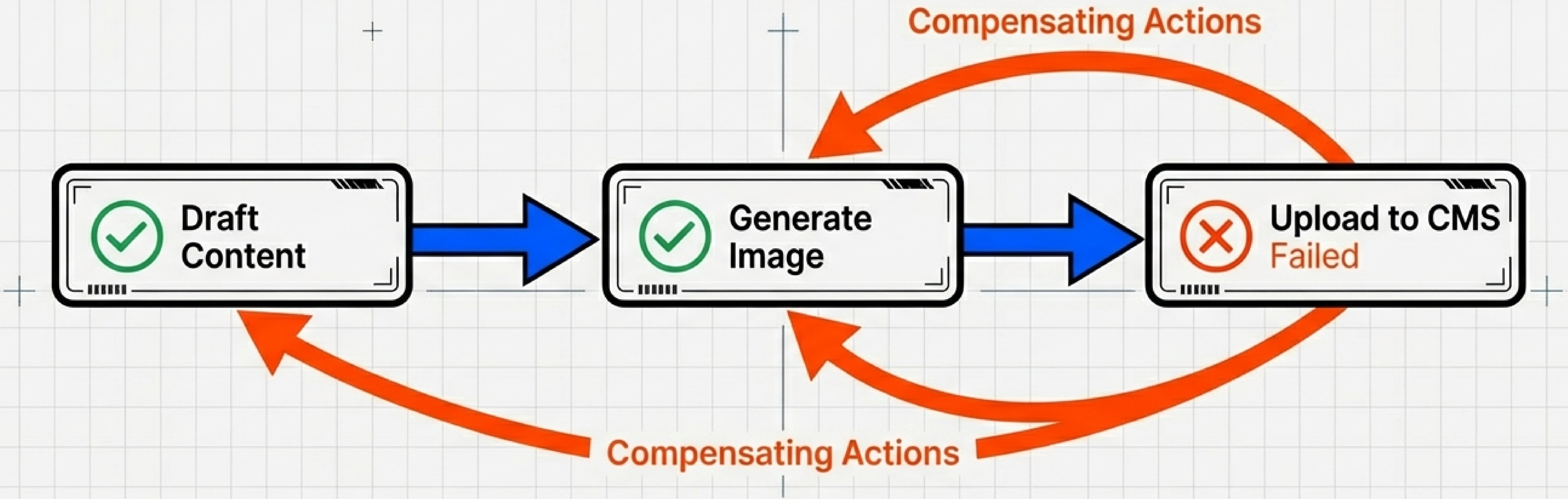
Outcome: System tunes its own instructions weekly based on real-world performance.

THE PROTOCOL LAYER: INTEROPERABILITY

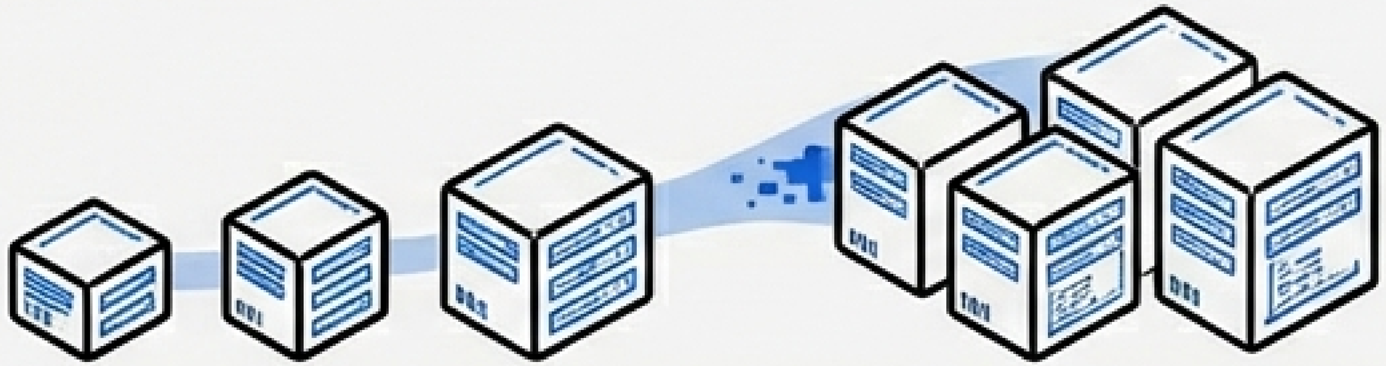


Standardized interfaces prevent vendor lock-in and enable dynamic team formation.

INFRASTRUCTURE & RESILIENCE: THE SAGA PATTERN



Elasticity Layer (Kubernetes)



JetBrains Nono

Horizontal Pod Autoscaler (HPA) monitors NATS queue depth. Automatically spins up “Friday-Dev” pods during spikes.

THE ROADMAP: AVOIDING THE “AGENTGPT ERROR”

PHASE 0: THE MANDATE

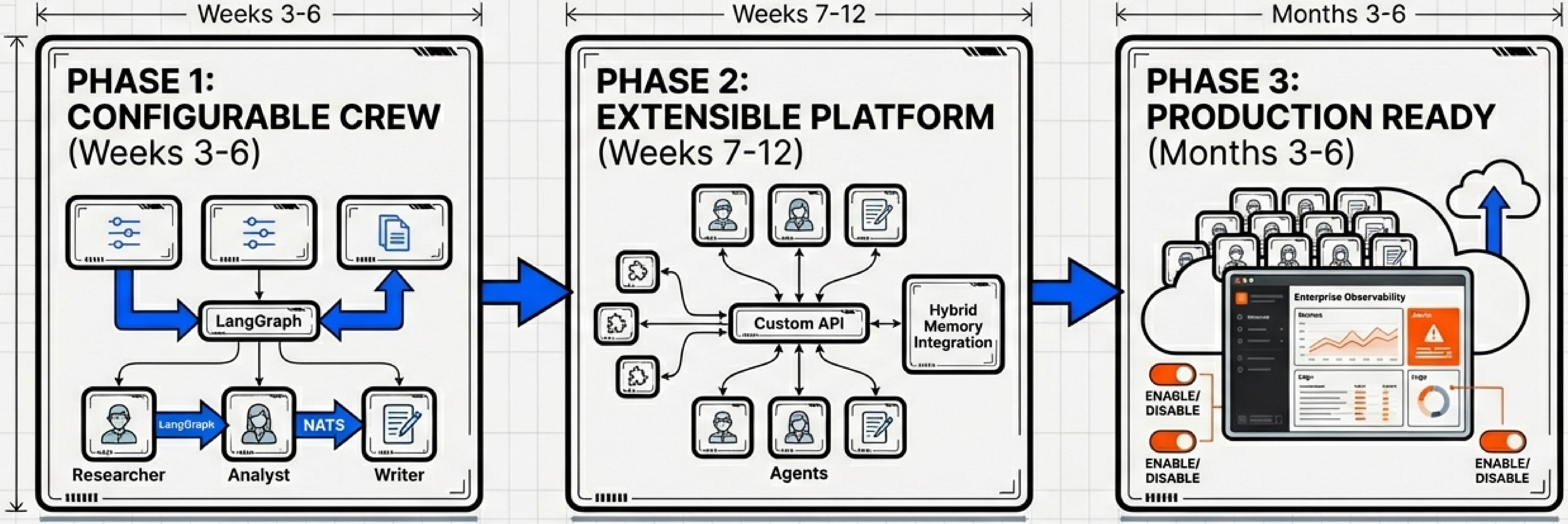
Ship 3 Agents doing 1 Task.

- `docBrain compose up` → Working Demo in < 5 Minutes.
- **Constraint:** No complex config. Hardcoded roles (Researcher, Analyst, Writer).

→ PHASE 1: SCALE → PHASE 2: AUTONOMY → PHAS PEOP

The Trap: Building a massive swarm immediately leads to high cash burn and low value. If 3 agents can’t collaborate reliably, 20 agents are just a faster way to burn money.

SCALING TO PRODUCTION



PHASE 1:
CONFIGURABLE CREW
(Weeks 3-6)
YAML-based definitions.
LangGraph Orchestration.
Basic NATS messaging.

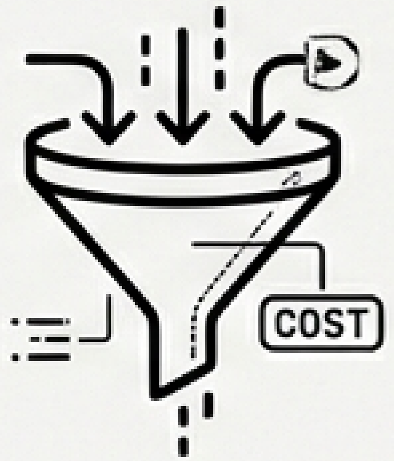
PHASE 2:
EXTENSIBLE PLATFORM
(Weeks 7-12)
Custom API. Plugin Architecture.
10-15 Agents.
Hybrid Memory Integration.

PHASE 3:
PRODUCTION READY
(Months 3-6)
Full 20+ Roster. Enterprise
Observability. Cloud Hosting.
Enable/Disable Toggles.

CORE PRINCIPLES OF SAMAS

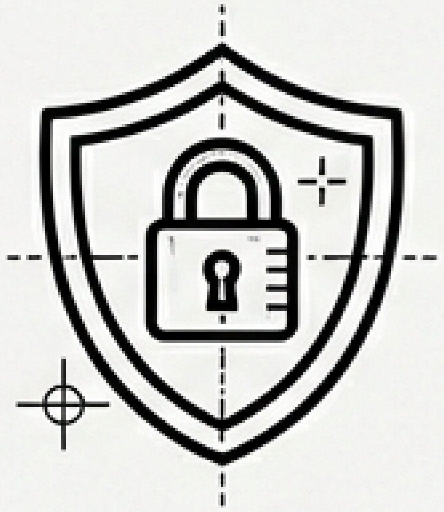
Orchestrated Simplicity

Staggered Cron > Websockets.



Cost Aware

Routing is fundamental architecture.



Trust but Verify

Confidence-weighted voting + Rule of Two.

Resilience

Event-driven recovery over raw autonomy.

SAMAS is not just about autonomy; it is about resilience in the real world.